

TEMASEK JUNIOR COLLEGE
2023 JC2 PRELIMINARY EXAMINATION
Higher 2



CANDIDATE
NAME

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CIVICS GROUP

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CENTRE
NUMBER

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INDEX
NUMBER

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Chemistry

9729/04

Paper 4 Practical

28 August 2023

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your centre number, index number, name and CG on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

Shift
Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 19 and 20.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	/ 55

This document consists of **17** printed pages and **3** blank pages.

Answer **all** the questions in the spaces provided.

- 1** In this experiment you are to determine the relative formula mass of an iron(II) salt by titration with potassium manganate(VII).

FA 1 is the iron (II) salt.

FA 2 is $0.0100 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .

FA 3 is 1.0 mol dm^{-3} sulfuric acid, H_2SO_4 .

Method

(a) Preparing a solution of **FA 1**

1. Weigh accurately about 4.0 g of **FA 1** into the plastic vial labelled **FA 1** and record the mass in the space below.
2. Transfer all the weighed **FA 1** into a 250 cm^3 beaker.
3. Use a measuring cylinder to add approximately 100 cm^3 of **FA 3** to the beaker. Stir until all the solid has dissolved.
4. Transfer the solution into the 250 cm^3 volumetric (graduated) flask labelled **FA 4**.
5. Wash out the beaker thoroughly using distilled water and add the washings to the volumetric flask. Make the solution up to the mark using distilled water.
6. Shake the flask thoroughly to mix the solution before using it for your titrations. Label this solution of the iron(II) salt as **FA 4**.

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[2]

(b) **Titration of FA 4 against FA 2**

1. Fill the burette with **FA 2**.

Retain the burette containing FA 2 for Q2.

2. Pipette 25.0 cm³ of **FA 4** into a conical flask.
3. Use a measuring cylinder to add 20 cm³ of **FA 3** to the flask.
4. Titrate **FA 4** with **FA 2** until the solution changes to a permanent pink colour.
5. Record your titration results, to an appropriate level of precision, in the space provided below.
6. Repeat steps 2 to 5 until consistent results are obtained.

(i) **Titration results**

[2]

- (ii) From your accurate titration results, obtain a suitable value to be used in your calculations.

Show clearly how you have obtained this value.

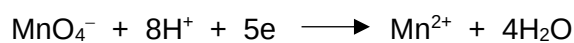
volume of **FA 2** = [4]

(c) Calculations

- (i) Calculate the number of moles of potassium manganate(VII) present in the volume of **FA 2** calculated in **(b)(ii)**.

moles of $\text{KMnO}_4 = \dots\dots\dots$ [1]

- (ii) The half-equation for the reduction of a manganate(VII) ion is:



Write the complete balanced ionic equation for the oxidation of iron(II) ion to iron(III) by manganate(VII) ions.

[1]

- (iii) Calculate the number of moles of iron(II) ions present in 25.0 cm^3 of solution **FA 4**.

moles of Fe^{2+} in 25.0 cm^3 of **FA 4** = $\dots\dots\dots$ [1]

- (iv) Calculate the number of moles of iron(II) ions present in 250 cm^3 of solution **FA 4**.

moles of Fe^{2+} in 250 cm^3 of **FA 4** = $\dots\dots\dots$ [1]

- (v) Calculate the relative formula mass of the iron(II) salt.

relative formula mass = [3]

- (d) A student conducted the above experiments and obtained an average titre value of 19.30 cm^3 .

The errors (uncertainties) associated with **each reading** using a graduated flask, pipette and burette are $\pm 0.15 \text{ cm}^3$, $\pm 0.10 \text{ cm}^3$ and $\pm 0.05 \text{ cm}^3$ respectively.

Calculate the maximum total percentage error (uncertainty) of this mean titre volume.

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..... [1]

[Total: 16]

Question 2 continues on the next page.

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- 2 Glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is a sugar that can act as a reducing agent. You will investigate how an increase in temperature affects the rate of the redox reaction between glucose and acidified potassium manganate(VII).

FA 2 is $0.010 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .

FA 3 is 1.0 mol dm^{-3} sulfuric acid, H_2SO_4 .

FA 5 is an aqueous solution containing 32.8 g dm^{-3} glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.

distilled water

You will measure the time it takes for the purple colour to disappear. Your table of results on page 8 should include the rate of reaction for each experiment.

(a) Method

Experiment 1

1. Add 10.00 cm^3 of **FA 2** from the burette (in question 1) into the 250 cm^3 beaker.
2. Use the 100 cm^3 measuring cylinder to transfer 50.0 cm^3 of **FA 3** into the beaker containing **FA 2**.
3. Use the same measuring cylinder to transfer 50.0 cm^3 of distilled water into the same beaker.
4. Place the beaker on the tripod and heat its contents to between 65°C and 70°C .
5. While the solution in the beaker is heating pour 25.0 cm^3 of **FA 5** into the 25 cm^3 measuring cylinder.
6. When the temperature of the contents of the beaker has reached between 65°C and 70°C , turn off the Bunsen burner and **carefully** place the hot beaker onto the white tile.
7. Record the initial temperature, T_i , of the solution in the beaker.
8. Add the 25.0 cm^3 of **FA 5** and **immediately** start timing.
9. Stir the contents of the beaker once and stop timing as soon as the solution turns colourless. Record the time to the nearest second.
10. Record the final temperature, T_f , of the solution as soon as it is colourless.
11. Calculate and record the average temperature, T_{avg} , of the reaction mixture to one decimal place.
12. Repeat Experiment 1 at four different temperatures, between room temperature and 70°C , each time using a clean and dry beaker. Record all your results in a table on page 8.

Results

The rate of reaction can be calculated as shown.

$$\text{Rate} = \frac{1000}{\text{reaction time}}$$

Calculate the rate of reaction to 3 sf for each experiment and include this in your table.

[4]

- (b) Plot a graph of rate (y-axis) against average temperature (x-axis) on the grid on page 9. Select a scale on the x-axis to include an average temperature of 15.0 °C. Label any points you consider anomalous.

Draw a line of best fit and extrapolate it to 15.0 °C.

[4]

- (c) **Use your graph** to calculate the **time** to the nearest second that the reaction would have taken if the average temperature had been 52.5 °C. Show **on the grid** how you obtained your answer.

Time = s [2]

- (d) Explain, by referring to your graph or your table of results, how the rate of reaction is affected by an increase in temperature.

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[1]

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- (e) Calculate the maximum percentage error in the reaction time recorded for **Experiment 1**. Assume the error of the timer is ± 1 s.

maximum percentage error in **Experiment 1** = % [1]

- (f) Suggest **two** ways to improve the accuracy of the results for this investigation.

1:

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2:

.....

.....

[2]

[Total: 14]

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Question 3 continues on the next page.

3 Qualitative Analysis

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, **describe at an appropriate juncture of your observations.**

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

- (a) **FA 6** and **FA 7** are aqueous solutions of equal concentrations, in mol dm^{-3} . Each contains one anion and one cation. The cation is the same in both **FA 6** and **FA 7**.

Half-fill the 250 cm^3 beaker with water. Heat the water to about 80°C and then **turn off the Bunsen burner**. This is the hot water bath needed for the tests below.

To about 2 cm depth of aqueous silver nitrate in a test-tube, add a few drops of aqueous sodium hydroxide to give a grey / brown precipitate. Then add aqueous ammonia dropwise until the precipitate **just** disappears. This solution is Tollens' reagent and is needed in a test below.

- (i) Carry out the tests on separate samples of **FA 6** and **FA 7** and complete the table.

Precaution : FA 6 should be kept stoppered at all times.

<i>test</i>	<i>observations</i>	
	FA 6	FA 7
To a 1 cm depth of solution in a test-tube in a test-tube rack, add a spatula measure of sodium carbonate.		
To a 1 cm depth of solution in a test-tube, add a few drops of acidified potassium manganate(VII). Place the test-tube in the hot water bath.		
To a 1 cm depth of Tollens' reagent in a test-tube, add a few drops of solution. Place the test-tube in the hot water bath and leave for several minutes.		

[3]

- (ii) From your observations in (i), identify the cation present in **both FA 6** and **FA 7**.

cation

[1]

- (iii) From your observations in (i), what can be deduced about the anion present in **FA 6**?

.....[1]

- (iv) Place a 1 cm depth of **FA 6** and **FA 7** separately in two test-tubes.

Measure and record the temperature of the two solutions.

FA 6 °C

FA 7 °C

To each solution, add an approximately 2 cm length of magnesium ribbon. Measure and record the maximum temperature reached in each test-tube.

FA 6 + Mg °C

FA 7 + Mg °C

[2]

- (v) Explain why there is a difference in the temperature rise for the reactions of magnesium with solutions **FA 6** and **FA 7**.

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.....

..... [2]

- (b) **FA 8** is a solution containing a mixture of two cations from those listed on page 19.

You are to plan a series of experiments that will enable you to identify the cations present. You should then carry out your plan, record all the observations you made in a suitable table and identify the cations present.

Cations present are and

[5]

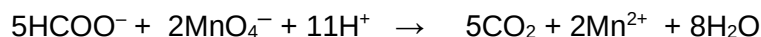
[Total: 14]

4 Planning

Magnesium methanoate, $\text{Mg}(\text{HCOO})_2$ has been used in studies to develop a multiresidue method for determining pesticides in fruits and vegetables.

A saturated solution of magnesium methanoate has a solubility of approximately 150 g dm^{-3} at room temperature. 50 cm^3 of the saturated solution was diluted to 250 cm^3 . This diluted solution is **FB 1**.

You are to conduct an experiment to determine the exact solubility of the saturated solution of magnesium methanoate by titrating against acidified potassium manganate (VII) as shown below. [M_r of $\text{Mg}(\text{HCOO})_2 = 114.3$]



You may assume you are provided with:

- 250 cm^3 of **FB1**, the diluted magnesium methanoate, $\text{Mg}(\text{HCOO})_2$, solution
- 0.02 mol dm^{-3} aqueous potassium manganate (VII), KMnO_4
- 100 cm^3 of 1.0 mol dm^{-3} sulfuric acid, H_2SO_4
- 250 cm^3 volumetric flask
- the equipment normally found in a school or college laboratory

- (a) Verify that the approximate concentration of HCOO^- in **FB1** is $0.525 \text{ mol dm}^{-3}$.

[1]

- (b) The concentration of HCOO^- in part (a) is unsuitable for titration with KMnO_4 .

Dilution is required in the volumetric procedure in order to obtain a reliable titre value.

Calculate a suitable volume of **FB 1** to be further diluted taking into account the equipment found in the college laboratory.

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[2]

- Your plan should include details of:

- then apparatus you would use
- the procedure you would follow
- the measurements you would take

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[4]

(d) A student diluted $x \text{ cm}^3$ of **FB 1** to 250 cm^3 and obtained $y \text{ cm}^3$ of KMnO_4 for the titration.

(i) Calculate the concentration of HCOO^- in **FB 1**.

[2]

(ii) Hence, calculate the concentration of HCOO^- in the saturated solution of magnesium methanoate.

[1]

(iii) Determine the solubility product of magnesium methanoate.

[1]

[Total: 11]

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Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

<i>cation</i>	<i>reaction with</i>	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

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(b) Reactions of aqueous anions

<i>anion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple